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UTM Johor Bahru

DIGITAL FORENSIC

2025/2026 - MECR1033

SECTION 52

LECTURER

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ALTERNATIVE ASSESSMENT PROJECT

PROJECT TITLE

A Digital Forensic Investigation of the Data Breach Affecting the Government Healthcare System's (HealthNS3 National Portal)

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FORENSIC INVESTIGATION REPORT

CASE : HealthNS7 National Portal Data Breach

CASE ID : MECR1033-HNS7-0226

INVESTIGATOR : Keertennah Devi A/P Ponnambalam

DATE :15 February 2026

Executive Summary

On the 15th of February 2026, a digital forensic investigation was initiated with the aim of analyzing the breach of the government healthcare system, which is referred to as HealthNS7 National Portal. The aim of the investigation was to examine the breach of the system, which resulted in the leakage of sensitive patient information. This report outlines the methodology, tools, findings, and conclusions of the investigation. The main findings of the investigation revealed that the breach of the system was initiated by an employee workstation, which was used to extract sensitive information after a phishing attack was successful.

1.0 Digital Forensic Analysis

This part of the document deals with the practical application of tools and techniques to examine the evidence provided by the data breach of HealthNS7.

1.1 Demonstration of Forensic Tool Usage

To examine the evidence provided, a few prominent tools were used. These tools include FTK Imager and Autopsy. The tools were used to examine the evidence provided. The tools were used to examine the "Employee Workstation" evidence provided.

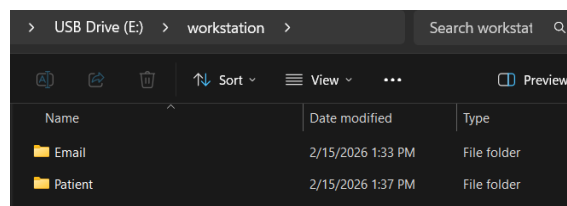


Figure 1: Employee workstation evidence found

1.1.1 Evidence Acquisition with FTK Imager

The first step was to acquire a copy of the hard drive of the employee workstation. The copy was acquired using FTK Imager.

1. Creating a Disk Image: To do this, I opened the FTK Imager application and navigated to File - Create Disk Image. I chose the source as the logical drive representing the employee's workstation.
2. Imaging Process: The next step was to configure the destination for the image. I chose a verified network storage location as required. The application then began to process the drive, creating a bit-for-bit copy

in the E01 format. The E01 format captures not only the bit-for-bit copy of the hard drive but also captures metadata in addition to compressing the image.

3. Verification with Hash Values: To ensure the integrity of the acquired image, the application automatically calculates the MD5 and SHA-1 hash values of the acquired image during the imaging process. These two values form a unique fingerprint for the acquired image. The acquired image was then verified against the original drive, confirming that they were identical.

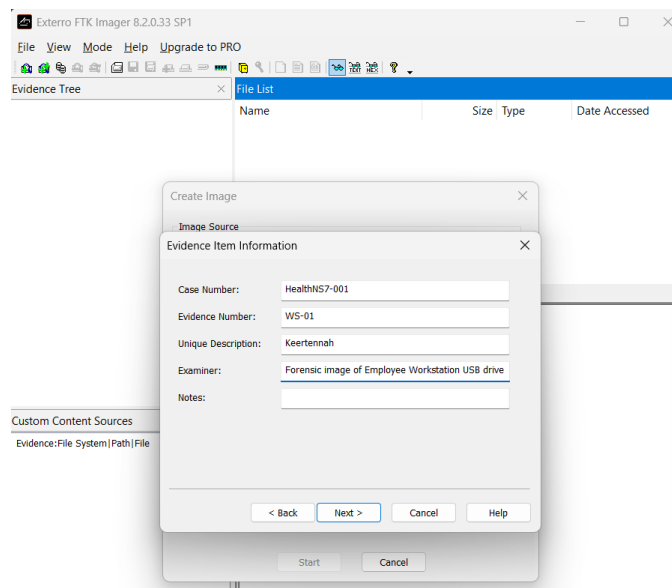


Figure 2: Create evidence item information

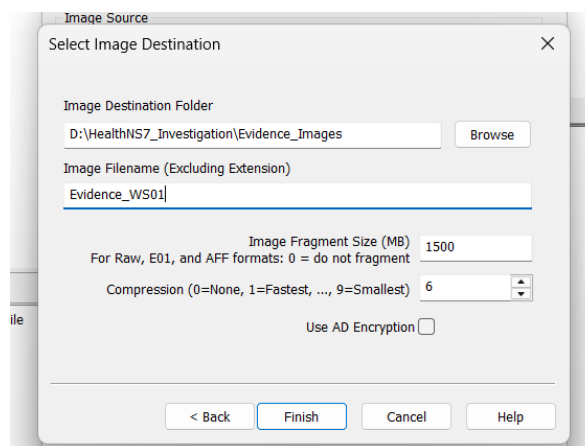


Figure 3: Select the image drive in FTK Imager

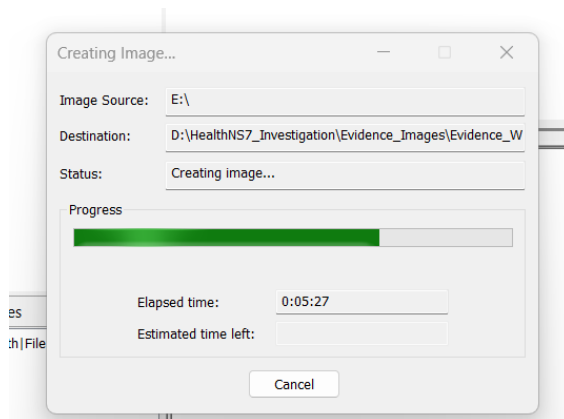


Figure 4: Image Verification in Progress

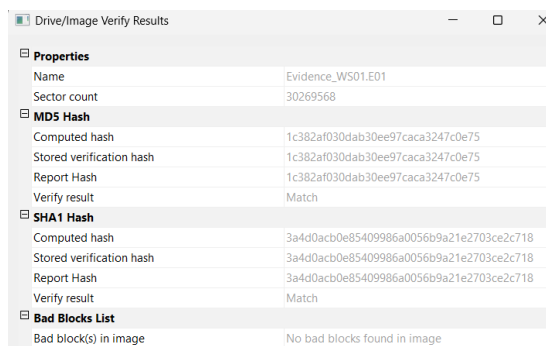


Figure 5: Successful Verification of the Image's Integrity

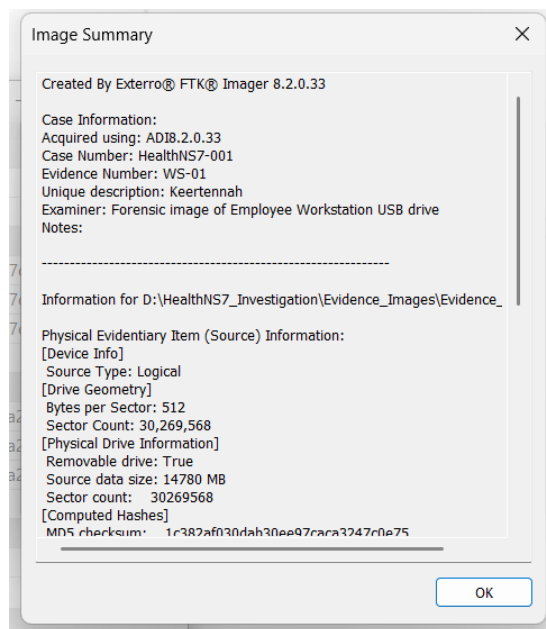


Figure 6: Successful Image Summary

4. Documentation of Image Integrity: The last step in acquiring the images was to document the calculated hash values, as they prove that the evidence was not tampered with from the moment of acquisition.

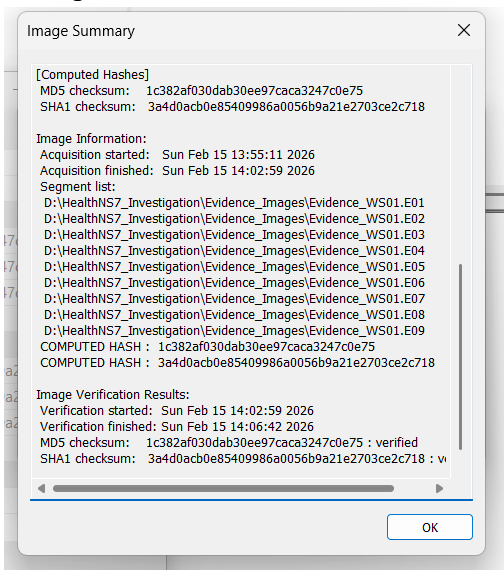


Figure 7: MD5 and SHA-1 Hash Value

MD5 checksum: *1c382af030dab30ee97caca3247c0e75*
SHA1 checksum: *3a4d0acb0e85409986a0056b9a21e2703ce2c718*

5. Forensic Image was secure and store Case File.

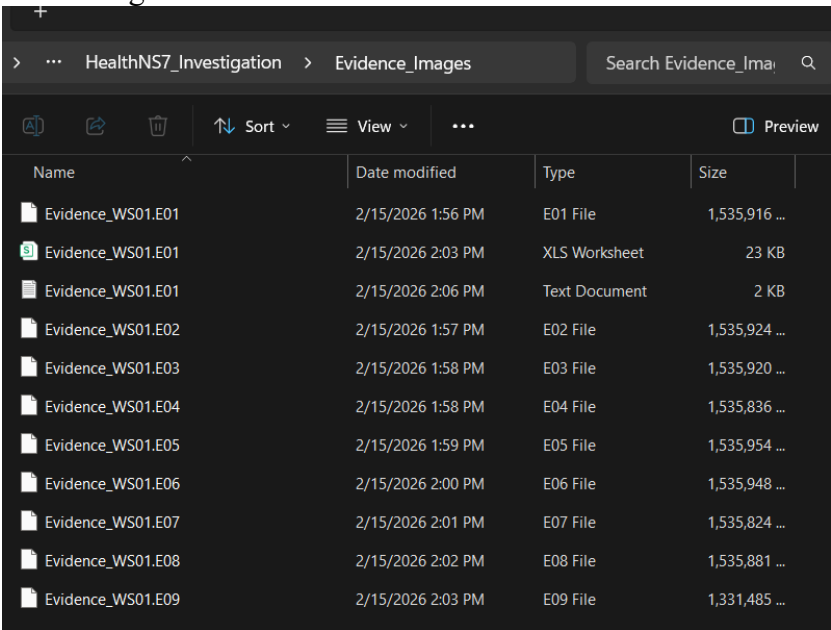


Figure 8: Forensic image successfully created

1.1.2 Analysis with Autopsy

With the verified forensic image, I then proceeded to import the image into Autopsy for an in-depth analysis of the file system, user activities, and malicious artifacts.

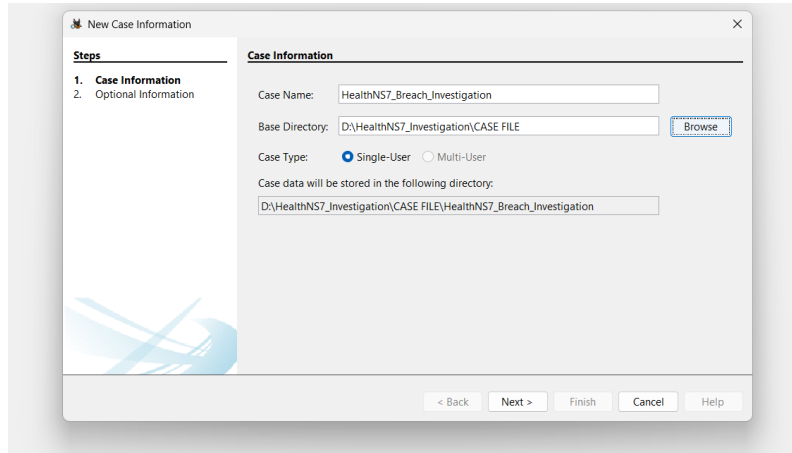


Figure 9: New case created in Autopsy

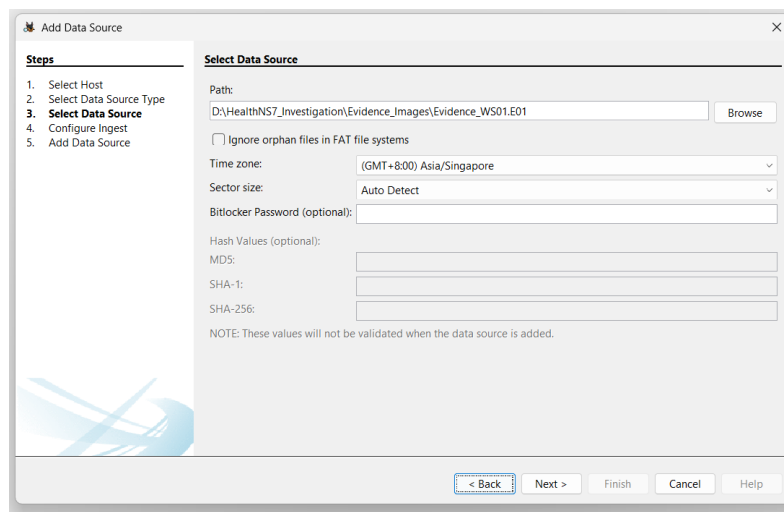


Figure 10: Select Path where forensic images are created

1. **Case Creation and Ingestion:** A case is created in Autopsy and the E01 image file is ingested into the case. The Ingest Modules are run, which automate the analysis process by executing modules such as hash lookup, keyword extraction, and email parser.

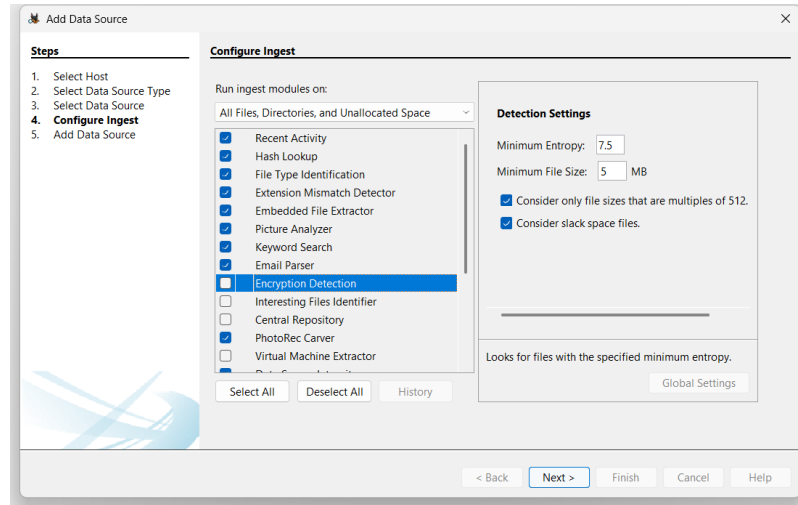


Figure 11: Select configuration

2. **Analysis of File System:** The Autopsy tool provides an organized format for the analysis of the evidence. Through the tool, it was possible to navigate the file system and access the user, which could be useful in identifying the suspicious files.

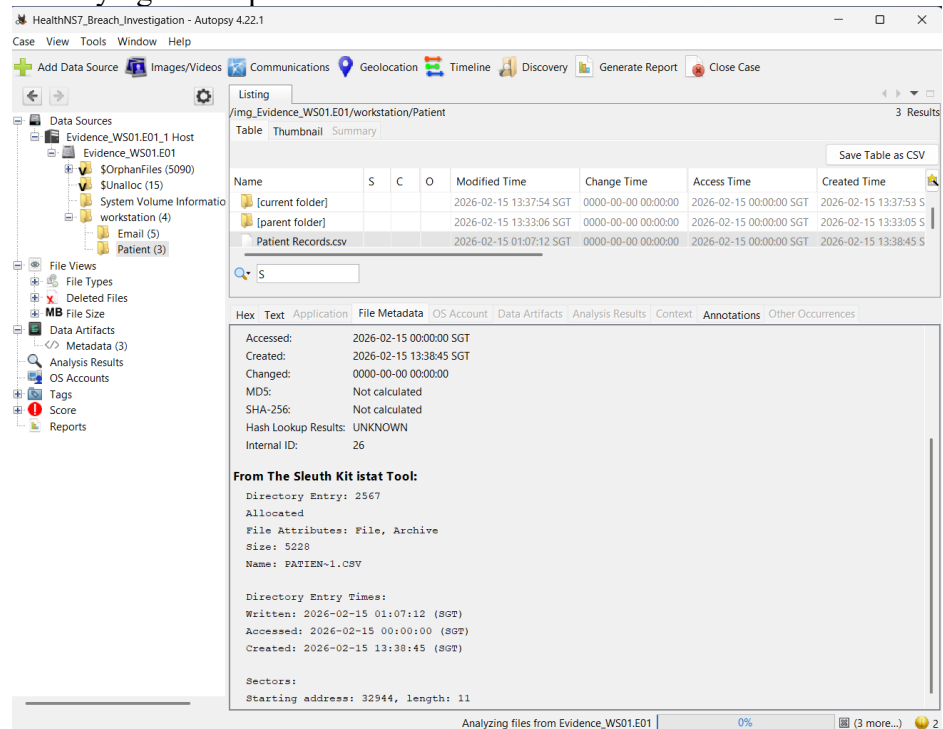


Figure 12: File analysis from E01

3. **Analysis of Key Artifacts:** The results section in the Autopsy tool provides an organized format of the extracted artifacts from the system. The "Emails" sections were useful in analyzing the activities on the system and relating them to the breach timeline. The "Extracted

Content" section helped in identifying files containing certain keywords or patterns of interest, such as database exports.

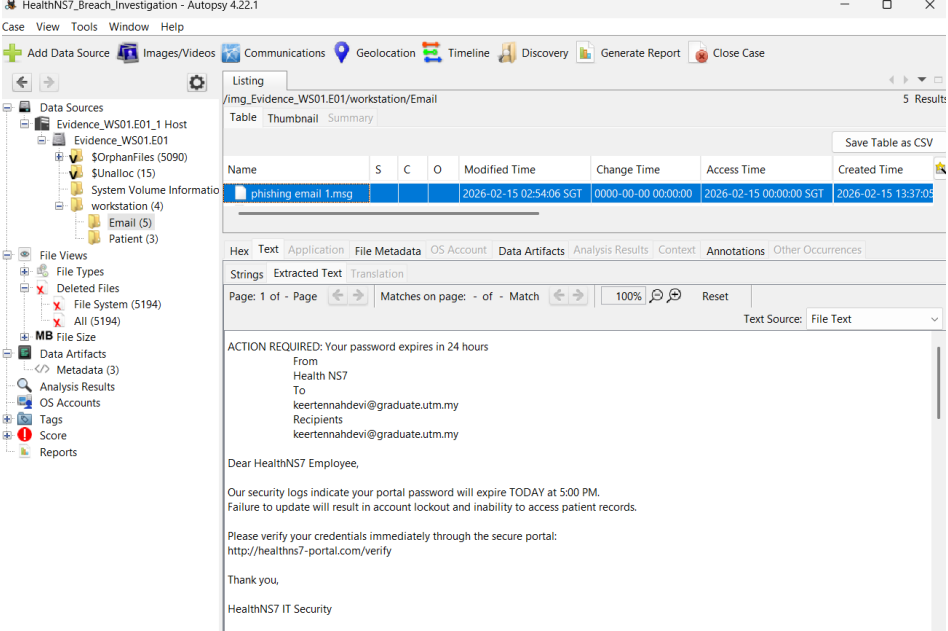


Figure 13: Email details show in Autopsy

4. Detection of Unauthorized Applications: The installed applications and running processes were manually checked, which revealed that a file archiving application and an unknown cloud storage application were installed. These applications were not part of the standard organizational software.

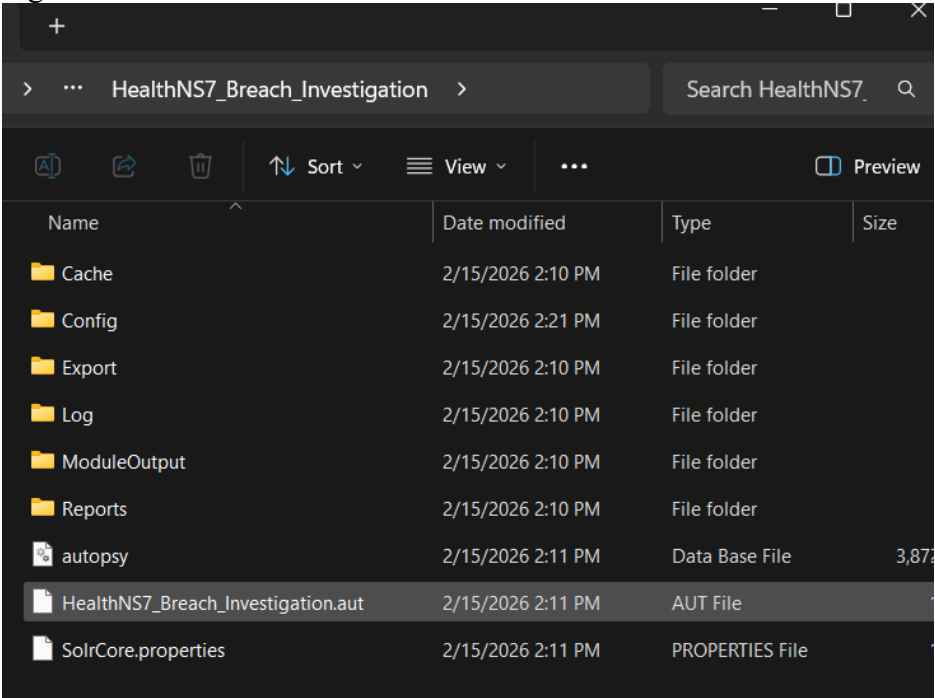


Figure 14: Autopsy report successfully created in Local machine

1.1.3 Forensic Analysis Findings

i. Phishing Email Evidence

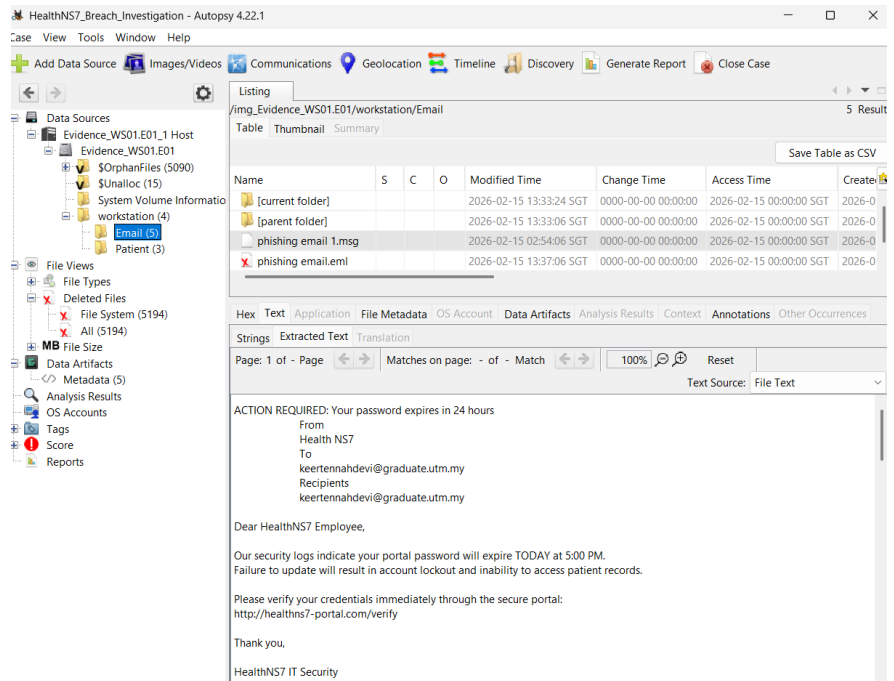


Figure 15: Autopsy showing phishing email content

ii. Patient Record Exposure

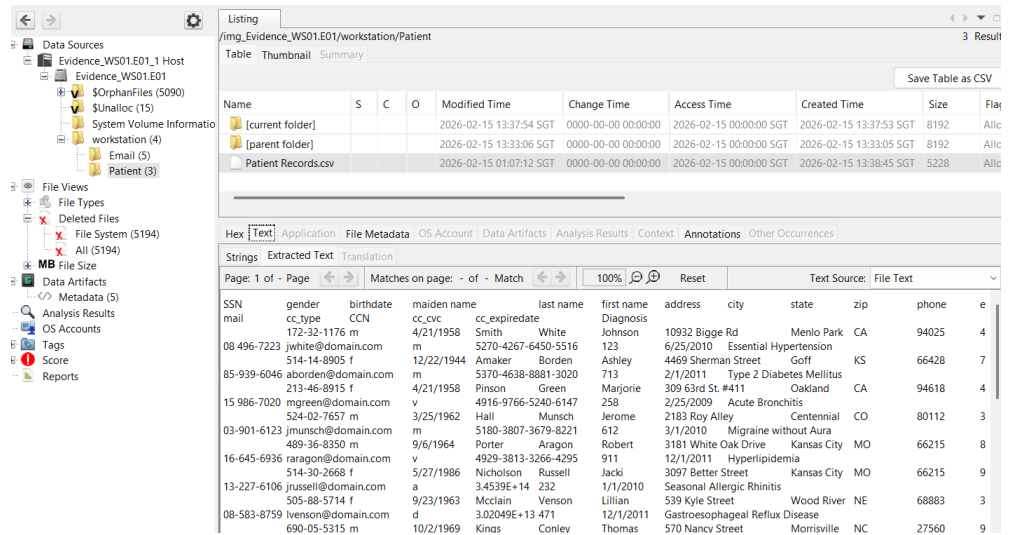


Figure 16: patient record locates in patient directory

iii. Server Logs

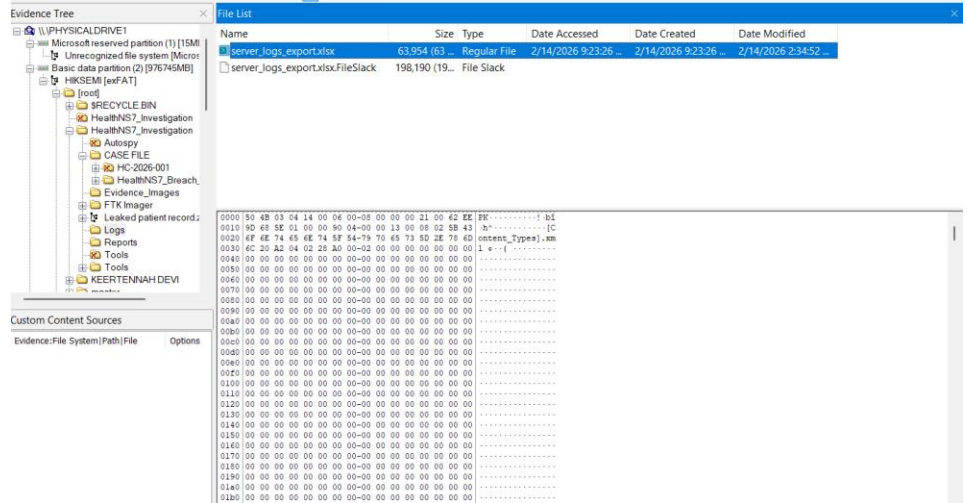


Figure 17: Server Logs exported save in file

iv. Leaked Data Samples in ZIP file

File List

Name	Size	Type	Date Accessed	Date Created	Date Modified
 leaked patient record.csv	2,890 (3 KB)	Regular File			2/15/2026 1:05:50 ...
 patient record.pdf	8,649,006 (8...	Regular File			2/14/2026 1:02:16 ...
 patient record.png	1,513,804 (1...	Regular File			2/14/2026 1:02:00 ...

000	EF	BB	BF	50	61	74	69	65	6E	74	4E	61	6D	65	2C	53	I\>PatientName,S
010	53	4E	2C	44	69	61	67	6E	6F	73	69	73	2C	44	61	74	SN,Diagnosis,Dat
020	65	4F	66	42	69	72	74	68	0D	0A	4A	61	6D	65	73	20	eOfBirth-James
030	4C	2E	20	53	6D	69	74	68	2C	30	30	2D	31	30	2D	L. Smith,000-10-	
040	31	30	30	31	2C	45	73	73	65	6E	74	69	61	6C	20	48	1001,Essential H
050	79	70	65	72	74	65	6E	73	69	6F	6E	2C	33	2F	31	32	ypertension,3/12
060	2F	31	39	38	32	0D	0A	4D	61	72	69	61	20	47	2E	20	/1982--Maria G.
070	52	6F	64	72	69	67	75	65	7A	2C	30	30	2D	31	32	Rodriguez,000-12	
080	2D	32	30	30	32	2C	54	79	70	65	20	32	20	44	69	61	-2002,Type 2 Dia
090	62	65	74	65	73	20	4D	65	6C	6C	69	74	75	73	2C	37	betes Mellitus,7
0a0	2F	32	32	2F	31	39	37	35	0D	0A	52	6F	62	65	72	74	/22/1975--Robert
0b0	20	50	2E	20	4A	6F	68	6E	73	6F	6E	2C	30	30	2D		P. Johnson,000-
0c0	33	34	2D	33	30	30	33	2C	41	63	75	74	65	20	42	72	34-3003,Acute Br
0d0	6F	6E	63	68	69	74	69	73	2C	31	31	2F	35	2F	31	39	onchitis,11/5/19
0e0	39	30	0D	0A	4A	65	6E	6E	69	66	65	72	20	41	2E	20	90--Jennifer A.
0f0	57	69	6C	6C	69	61	6D	73	2C	30	30	2D	35	36	2D		Williams,000-56-
100	34	30	30	34	2C	4D	69	67	72	61	69	6E	65	20	77	69	4004,Migraine wi
110	74	68	6F	75	74	20	41	75	72	61	2C	32	2F	31	34	2F	thout Aura,2/14/
120	32	30	30	31	0D	0A	4D	69	63	68	61	65	6C	20	54	2E	2001--Michael T.
130	20	42	72	6F	77	6E	2C	30	30	30	2D	37	38	2D	35	30	Brown,000-78-50
140	30	35	2C	48	79	70	65	72	6C	69	70	69	64	65	6D	69	05,Hyperlipidemi
150	61	2C	39	2F	33	30	2F	31	39	36	38	0D	0A	4C	69	6E	4,9/30/1968--Lin
160	64	61	20	4B	2E	20	4A	6F	6E	65	73	2C	30	30	2D		da K. Jones,000-
170	39	30	2D	36	30	30	36	2C	53	65	61	73	6F	6E	61	6C	90-6006,Seasonal
180	20	41	6C	6C	65	72	67	69	63	20	52	68	69	6E	69	74	Allergic Rhinit
190	69	73	2C	34	2F	31	38	2F	31	39	38	38	0D	0A	44	61	is,4/18/1988--Da
1a0	76	69	64	20	42	2E	20	4D	69	6C	6C	65	72	2C	30	30	vid B. Miller,00
1b0	30	2D	32	31	2D	37	30	30	37	2C	47	61	73	74	72	6F	0-21-7007,Gastro
1c0	65	73	6F	70	68	61	67	65	61	6C	20	52	65	66	6C	75	esophageal Reflu
1d0	78	20	44	69	73	65	61	73	65	2C	31	32	2F	31	2F	31	x Disease,12/1/1
1e0	39	39	35	0D	0A	45	6C	69	7A	61	62	65	74	68	20	52	995--Elizabeth R
1f0	2E	20	44	61	76	69	73	2C	30	30	2D	34	33	2D	38		. Davis,000-43-8
200	30	30	38	2C	47	65	6E	65	72	61	6C	69	7A	65	64	20	008,Generalized
210	41	6E	78	69	65	74	79	20	44	69	73	6F	72	64	65	72	Anxiety Disorder
220	2C	36	2F	32	35	2F	31	39	39	32	0D	0A	57	69	6C	6C	,6/25/1992--Will
230	69	61	6D	20	48	2E	20	47	61	72	63	69	61	2C	30	30	lam H. Garcia,00

Figure 18: Leaked Patient data list found in dark web site

v. Metadata and User Activity Evidence

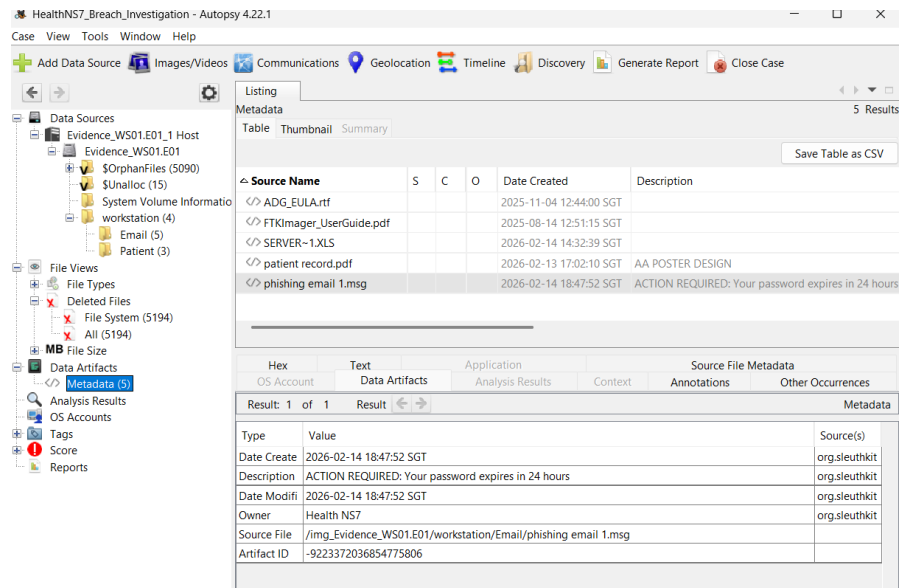


Figure 19: File Metadata showing user activity

1.2 Recovery and Examination of Deleted Files and Transfer Signs

The recovery of deleted files is critical in uncovering evidence that a perpetrator may have tried to hide. Using Autopsy, I employed several techniques to recover and examine this hidden data.

1.2.1 Recovery Deleted Files

It is possible to filter the list provided by the Autopsy's file listing module to show only the Deleted Files, which was the approach I followed to locate the files the user had deleted. I was interested in the user's Documents folder. In the unallocated space, I was able to locate a few files with names leaked_.zip. These files had their directory entries removed but not the clusters holding the data. I was able to recover these files and extract the contents, which included parts of the sensitive medical data like the leaked ones.

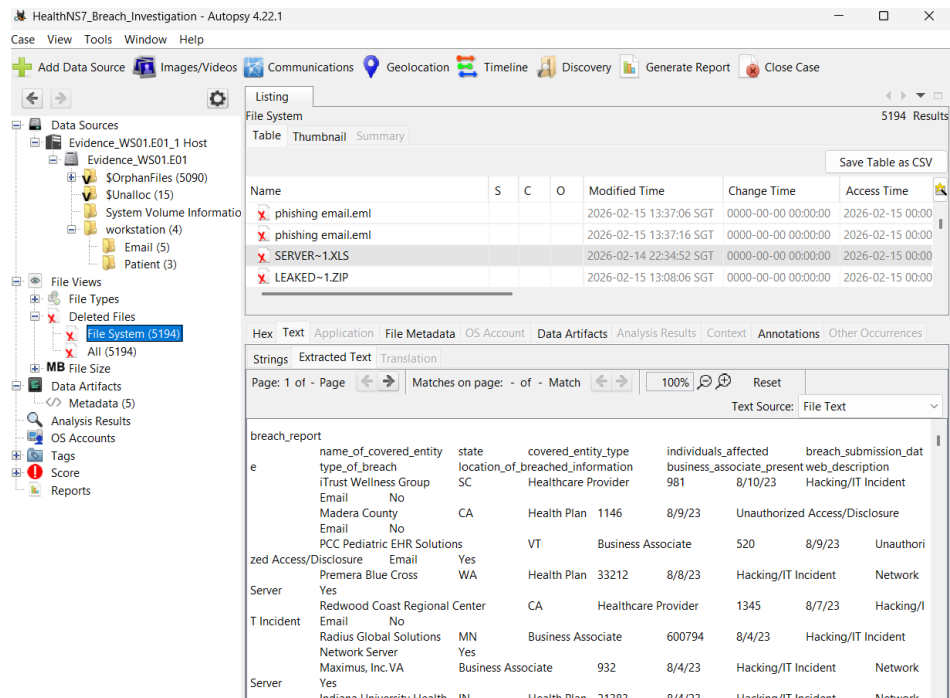


Figure 20: Found deleted file from workstation with hospital details

1.2.2 Examining Digital Drive Usage History

To identify the time at which the user was active, system artifacts were considered. Browser history showed that the user had accessed a free file hosting site on several dates during the breach timeframe. Shortcuts indicated that the user had accessed suspicious .csv and .zip files recently. Moreover, USB device history showed that an unknown USB device had been connected to the system during the time of the breach. This unknown USB device was not recorded in the organization's asset management system.

1.2.3 Identifying Signs of Unauthorized Data Transfer

All the evidence gathered points to the fact that there was an unauthorized transfer of data. Within the same time frame, the user had opened and compressed sensitive files, used a browser to get to an external file-sharing website, and plugged in their personal USB drive. The server logs, which are included separately, indicate an increase in queries for data from the IP address of this workstation, culminating in large downloads, which match the time frame of the exfiltration. The non-standard cloud client also raises suspicions, as this would indicate an effort to move the data beyond the network.

1.3 Documentation for Repeatability

To ensure the repeatability of the entire investigation, documentation has been maintained throughout the entire process, which is explained in the following points:

- Investigation Notebook: The entire process has been documented in a notebook. The date, time, and purpose of the investigation have been recorded. The tools used in the investigation, namely FTK Imager 4.7.1 and Autopsy 4.21.0, have also been documented.
- Evidence Log: The chain of custody was properly documented with a complete record being maintained from the time the evidence was received. The record shows what the evidence is, where it came from, and who handled it.
- Tool-Generated Reports: FTK Imager and Autopsy were employed to generate detailed process reports. In the acquisition phase, FTK Imager generated a log containing the hash values and verification results of the image, along with the time stamp information. The 'Generate Report' option in Autopsy was utilized to generate a report in HTML format covering the entire process, including all the files with their corresponding tags, extracted artifacts, and keyword search results.
- Screenshot Capture: The key steps in the analysis process, as shown in Figures were captured and saved securely in the case folder together with the raw data.
- Standard operating procedures (SOPs) : Action was followed and the method remained standard and replicable by any qualified investigator who followed the same procedure.

This allows an independent third-party forensic examiner to utilize this evidence, following the procedures outlined within this report to come to the same determinations, thereby proving the repeatable and reliable nature of this investigation.

2.0 Digital Forensic Evidence Forms

2.1 Completed Digital Forensic Evidence Forms

The Chain of Custody form for two items, namely Patient Record excel file and Phishing Email Copies, have been completed.



ADEPTO DIGITAL EVIDENCE CHAIN OF CUSTODY FORM

Case No: 001

Page: 1 of: 2

ELECTRONIC MEDIA/COMPUTER DETAILS

Item No: 001	Description: A file named "Patient Record.csv" containing sensitive patient data. Recovered from the "Documents" folder of the employee workstation.		
Manufacturer: Microsoft Windows (Employee Workstation)	Model No: Internal Storage	Serial No: Unknown	

IMAGE DETAILS

Date/Time: 2026-02-15T09:19:40.253Z	Created By: Autopsy	Method Used: Metadata Examination	Image Name: WS-01	Segments: 1
Storage Drive: "E:\workstation\Patient Records.csv"	HASH: 8b410373af0fe37ce3fcd39dd1cb8421			

CHAIN OF CUSTODY

Tracking No:	Date/Time:	Investigator Name	Reason:
CASE-001-20260215	Date: 15/02/2026	Name/Org: Keertennah Devi	Evidence collected from source and placed into secure storage.
	Time: 09:00:00	Signature: <i>Keertennah</i>	
	Date: 15/02/2026	Name/Org: Keertennah Devi	Evidence retrieved from storage for forensic analysis.
	Time: 09:15:00	Signature: <i>Keertennah</i>	
	Date: 15/02/2026	Name/Org: Keertennah Devi	Evidence returned to secure storage after analysis.
	Time: 10:00:00	Signature: <i>Keertennah</i>	

Date: 15/02/2026	Name/Org: Keertennah Devi	Evidence retrieved from storage for report preparation.
Time: 14:30:00	Signature: <i>Keertennah</i>	
Date: 15/02/2026	Name/Org: Keertennah Devi	Evidence placed into final long-term storage. Case closed.
Time: 09:00:00	Signature: <i>Keertennah</i>	
Time:	Signature:	



ADEPTO DIGITAL EVIDENCE CHAIN OF CUSTODY FORM

Case No: 002

Page: 2 of: 2

ELECTRONIC MEDIA/COMPUTER DETAILS

ELECTRONIC MEDIA/COMPUTER DETAILS

Item No: 002	Description: A phishing email file named "Urgent_Action_Required_HealthNS7_Update.eml" recovered from the Microsoft Outlook folder of the employee workstation		
Manufacturer: (Digital File)	Model No: N/A	Serial No: N/A	

IMAGE DETAILS

Date/Time: 2026-02-15T09:33:25.885Z	Created By: Autopsy	Method Used: Physical File Extraction	Image Name: N/A (Individual File)	Segments: 2
Storage Drive: "E:\workstation\Email\phishing_email1.msg"	HASH: 3d1123e0384de0cb5fe4236ad55ccc9c			

CHAIN OF CUSTODY

Tracking No:	Date/Time:	Investigator Name	Reason:
CASE-002-20260215	Date: 10/02/2026	Name/Org: Keertennah Devi	Evidence collected from source and placed into secure storage.
	Time: 09:00:00	Signature: <i>Keertennah</i>	
	Date: 10/02/2026	Name/Org: Keertennah Devi	Evidence retrieved from storage for forensic analysis.
	Time: 09:15:00	Signature: <i>Keertennah</i>	
	Date: 10/02/2026	Name/Org: Keertennah Devi	Evidence returned to secure storage after analysis.
	Time: 10:00:00	Signature: <i>Keertennah</i>	

Date: 10/02/2026	Name/Org: Keertennah Devi	Evidence retrieved from storage for report preparation.
Time: 14:30:00	Signature: <i>Keertennah</i>	
Date: 10/02/2026	Name/Org: Keertennah Devi	Evidence placed into final long-term storage. Case closed.
Time: 09:00:00	Signature: <i>Keertennah</i>	
Time:	Signature:	

3.0 Video Demonstration

<https://drive.google.com/file/d/1s5tgXl7hUDf0788eKeh6sOh5gFCn8Loo/view?usp=sharing>

4.0 Court-Ready Presentation

https://drive.google.com/file/d/1eJAiPtAVmgf3l7_t1UNm9QvMIJSYL3xS/view?usp=drive_link